

Introduction to neural networks

Regression introduction

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Linear regression



Linear regression

- A relation between:
- Independent variables (regressors)
- Dependent variables (responses)

$$Y = \beta_0 + \beta_1 x \quad (1)$$

- β_0 is the intercept
- β_1 is the slope

Linear regression

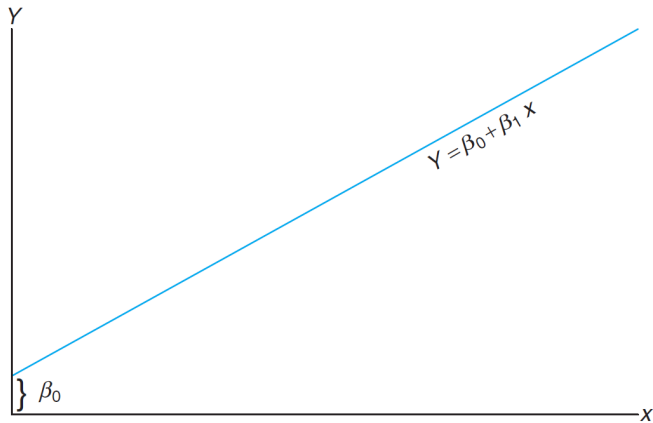


Figure: Deterministic equation for a linear model

Simple Linear Regression (SLR) Model

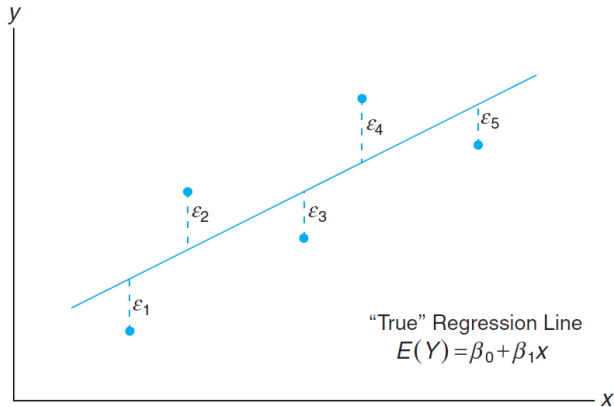
- SLR Model

$$Y = \beta_0 + \beta_1 x + \epsilon \quad (2)$$

- where $E(\epsilon) = 0$ and $Var(\epsilon) = \sigma^2$

Simple Linear Regression (SLR) Model

- What does it mean?

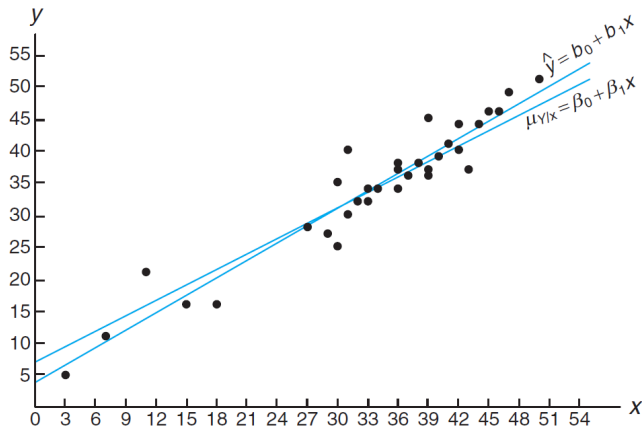


- To estimate the regression coefficients (β_0, β_1)

$$\hat{y} = b_0 + b_1x \quad (3)$$

Fitted regression line

- The fitted regression line and a hypothetical true regression line are shown on the scatter diagram



- Relation with respect of the "true" regression line.

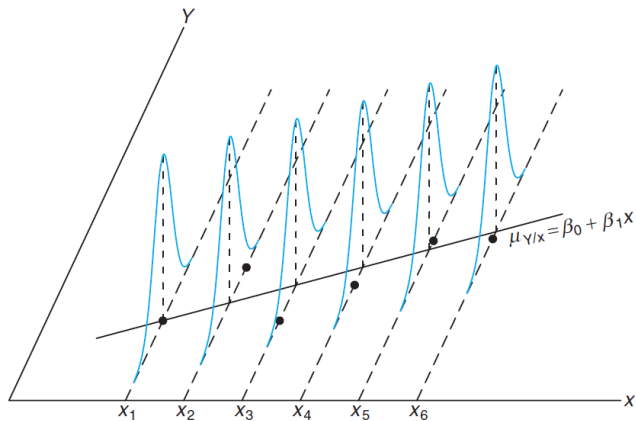


Figure: Individual observations around the true regression

- Error in fit: Given a set of regression data $\{(x_i, y_i) : i = 1, 2, \dots, n\}$ and a fitted model $\hat{y} = b_0 + b_1x$, the i -th residual, e_i , is given by

$$e_i = y_i - \hat{y}_i, i = 1, 2, \dots, n$$

- e_i is observable
- ϵ_i is not observable

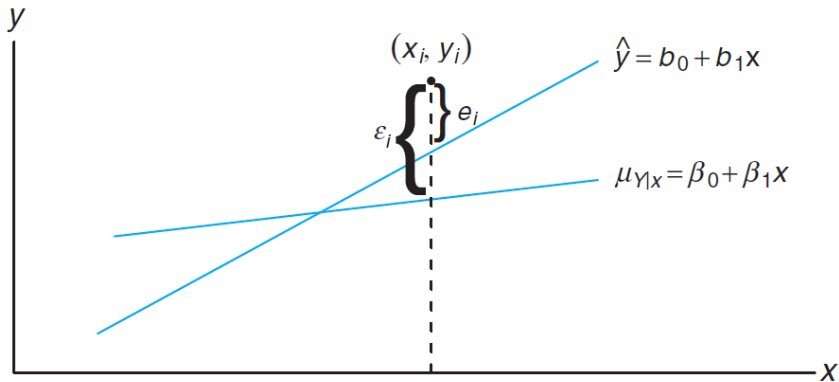


Figure: Comparing ϵ_i with the residual e_i

Least Squares

- We shall find b_0 and b_1 so that the sum of the squares of the residuals should be minimum.
- Sum of squares of the error

$$SSE = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2.$$

- The method for estimating the parameters is called **least squares**.

Differentiating SSE with respect to b_0 and b_1 , we have

$$\frac{\partial(SSE)}{\partial b_0} = -2 \sum_{i=1}^n (y_i - b_0 - b_1 x_i), \quad \frac{\partial(SSE)}{\partial b_1} = -2 \sum_{i=1}^n (y_i - b_0 - b_1 x_i) x_i.$$

Setting the partial derivatives equal to zero and rearranging the terms, we obtain the equations (called the **normal equations**)

$$nb_0 + b_1 \sum_{i=1}^n x_i = \sum_{i=1}^n y_i, \quad b_0 \sum_{i=1}^n x_i + b_1 \sum_{i=1}^n x_i^2 = \sum_{i=1}^n x_i y_i,$$

which may be solved simultaneously to yield computing formulas for b_0 and b_1 .

- Parameters:

Given the sample $\{(x_i, y_i); i = 1, 2, \dots, n\}$, the least squares estimates b_0 and b_1 of the regression coefficients β_0 and β_1 are computed from the formulas

$$b_1 = \frac{n \sum_{i=1}^n x_i y_i - \left(\sum_{i=1}^n x_i \right) \left(\sum_{i=1}^n y_i \right)}{n \sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i \right)^2} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \text{ and}$$
$$b_0 = \frac{\sum_{i=1}^n y_i - b_1 \sum_{i=1}^n x_i}{n} = \bar{y} - b_1 \bar{x}.$$

- **Ejercicio 11.1**, Walpole, R. E., Myers, R. H., Myers, S. L., & Ye, K. Probability and statistics for engineers and scientists (9th Ed). New York: Macmillan.

